BUSINESS MEMO: COD RTO PREVENTION STRATEGY

To: Founder, D2C Haircare Brand

From: Associate Software Engineer – Take-Home Assignment

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Subject: Reducing COD RTO While Protecting Genuine Customer Orders

1. Executive Summary

The supplied historical dataset initially contained **2,018 rows**. After removing 30 exact duplicate rows, 1,988 unique records remained. Of these, 1,987 had known delivery outcomes: 1,422 delivered and 565 returned to origin (RTO). The historical RTO rate is **28.43%**.

I developed a configurable COD decision engine that classifies orders into four actions: ship as-is, WhatsApp confirmation, prepaid nudge, and hold/cancel review. The recommendation is to intervene selectively rather than reject all COD orders. A hypothetical financial simulation estimates a net positive impact of ■**24,704.55** for the evaluated cohort. This is an assumption-based estimate, not realized savings.

2. Three Surprising Findings from the Data

Finding 1 – Previous RTO history is a strong risk signal.

Previous RTO count	Orders	RTO rate
0	1,845	27.15%
1	108	35.19%
2	34	76.47%

Customers with two previous RTOs had a 76.47% RTO rate. This supports prioritizing repeat-RTO customers for WhatsApp confirmation rather than treating all COD customers equally.

Finding 2 – Very short addresses are associated with much higher RTO.

Address length	Orders	RTO rate
0–15 characters	183	61.75%
16–30 characters	136	26.47%
31–45 characters	1,587	24.83%
46–60 characters	81	27.16%

Orders with addresses of 0–15 characters had a 61.75% RTO rate, compared with 24.83% for 31–45-character addresses. This supports address completeness checks and manual verification of suspiciously short addresses. Address length alone does not prove an address is invalid.

Finding 3 – RTO varies by marketing source.

Marketing source	Orders	RTO rate
Meta Ads	765	31.24%
Instagram Organic	232	29.31%
Google Ads	244	28.28%
Influencer	225	27.11%
WhatsApp Broadcast	268	26.12%
Direct	253	22.92%

Meta Ads contributed the highest order volume and had the highest RTO rate among these sources. This is a reason to monitor campaign quality and customer confirmation, not to block Meta Ads customers.

Additional Operational Observation

Xpressbees had a 30.95% RTO rate across 391 orders, compared with Delhivery's 26.04% across 699 orders. Courier performance should be monitored before changing allocation decisions.

3. Recommended Decision Rules

Decision	Orders	Share
Ship as-is	1,654	83.2%
WhatsApp confirmation	235	11.8%
Nudge to prepaid	99	5.0%
Hold/cancel review	0	0.0%

Recommended operating rules:
1. **Ship as-is:** Orders without significant risk indicators.
2. **WhatsApp confirmation:** Customers with repeated RTO history or suspiciously short/incomplete address details.
3. **Nudge to prepaid:** Eligible customers with one previous RTO, using a 5% discount and configurable minimum order value.
4. **Hold/cancel review:** Missing or clearly invalid addresses should be reviewed before dispatch. Avoid automatic cancellation based only on city, pincode, marketing source, or first-time customer status.

Thresholds are configurable without changing application code. Historical decision counts represent retrospective classification, not proof that interventions would have changed actual outcomes.

4. Operations Versus Marketing: Financial Trade-off

The estimated cost of one RTO is ■**169.40**: forward shipping ■65 + return shipping ■65 + packaging/handling ■25 + expected damage cost (8% × ■180 = ■14.40).

The simulation assumes 30% RTO prevention in the WhatsApp confirmation group, 5% false cancellations among delivered orders in that group, 20% prepaid acceptance, a 4% prepaid RTO rate, and a 5% prepaid discount.

Financial measure	Estimated amount
Potential RTO savings	■27,278.48
Lost genuine-customer contribution	■1,931.42
Prepaid discount cost	■642.51
Estimated net impact	**■24,704.55**

Operations benefits from fewer avoidable returns. Marketing is protected by using confirmation and incentives instead of blanket COD restrictions. This is particularly important when new-customer acquisition costs ■300 or more.

5. Risks and Monitoring

• Genuine customers may not respond: track WhatsApp response rates and manually review non-responses before cancellation.
• Incorrect address flags: review AI-flagged records and monitor false positives, especially for regional and Hinglish addresses.
• Discounts may reduce contribution: compare prepaid conversion value with discount expense.
• Courier and city differences may be misleading: monitor sample sizes and changing RTO rates before changing allocation or regional policies.
• LLM errors or API failure: keep deterministic rules and local validation as fallback. Use an LLM for ambiguous text, not financial calculations or independent cancellation decisions.
• Historical data may not represent current behavior: monitor new orders and evaluate decisions against actual delivery outcomes.

6. Next Two-Week Plan

Week 1 – Controlled testing
• Run the engine in shadow mode on new COD orders.
• Validate WhatsApp YES, NO, random reply, and 24-hour timeout flows.
• Manually review high-risk addresses and repeat-RTO customers.
• Compare decisions with actual delivery outcomes.

Week 2 – Pilot and measurement
• Run a limited WhatsApp confirmation and prepaid discount pilot.
• Measure confirmation response, prepaid conversion, RTO rate, false cancellation rate, and discount cost.
• Compare actual savings with lost contribution and acquisition cost.
• Tune thresholds before wider rollout.

Final Recommendation

Start with selective WhatsApp confirmation for high-risk orders and a 5% prepaid incentive for eligible customers. Do not impose blanket COD restrictions or automatically cancel first-time customers. Use the first two weeks to validate assumptions with real delivery outcomes before scaling.

Data and financial limitations: This analysis uses the supplied historical sample. The dataset is not a live order feed, and the available results do not establish a specific last-month cohort. City, courier, and marketing-source differences are observational associations. Financial impact figures are hypothetical scenario estimates, not guaranteed or realized savings.